

Dante Switch Runbook

Tech Table • FOH • BigRack • Attic | dark day 2026-07-14 | per-switch remediation from the config audit + Dante best-practice research (Audinate / Yamaha / Biamp / Shure)

Reference article: **Dante on Cisco Switches** in the KB (dante-cisco-switch-config). This runbook is the do-it list: exact commands per switch, in order. Every CLI block below is paste-ready from an SSH/console session in config mode, but each step also names the GUI path — the two firmware generations occasionally reject a keyword, and the GUI always works. Where a command is marked **VERIFY FIRST**, run the show command before changing anything: these files are backups and may be stale against the live switches.

Order matters: BigRack (core) gets the querier BEFORE snooping+filtering goes live on the edges — snooping with no querier kills Dante after ~5 minutes as group registrations age out. Work BigRack → FOH → Tech Table → Attic. Nothing here should drop audio on a dark day, but treat trunk edits as link-risk: have physical access to the switch you're editing.

Before anything: pull a fresh backup of all four running configs (Administration → File Management → Download/Backup Configuration, source = Running). After everything: save running→startup on all four (copy running-config startup-config) and pull new backups. Config on these switches is volatile until saved.

Target state

	BigRack .254 (core)	FOH .251	Tech Table .253	Attic .252
Platform / queues	SG500X-class / 4	CBS350-class / 8	CBS350-class / 8	SG500-class / 4
IGMP snooping VLAN 200	ON	ON	ON (has it)	ON (add all)
Bridge mcast filtering	ON (has it)	ON (add)	ON (has it)	ON (add)
IGMP querier VLAN 200	ON — the only one, v3	OFF (remove addr line)	OFF (demote)	OFF
Unregistered multicast	Forwarding	Forwarding	Forwarding	Forwarding
QoS	Basic, trust DSCP	Basic, trust DSCP	Basic, trust DSCP	Basic, trust DSCP
DSCP 56 / 46 / 8 queue	4 / 3 / 2	8 / 7 / 2	8 / 7 / 2	4 / 3 / 2
STP	RSTP, root (4096)	RSTP	RSTP	RSTP
Trunks	static, no Smartport	gi10 +100,200,300	already static	static, no Smartport

Why one querier on the core: group memberships need periodic queries to stay alive; one querier per VLAN, election left enabled (lowest IP wins if a rogue appears). Why unregistered multicast stays Forwarding: Dante discovery (mDNS 224.0.0.251) and control (224.0.0.230–233) are link-local and never IGMP-joined — filter them and devices vanish from Dante Controller. VLAN 300 (sACN) stays UN-snooped: sACN nodes that don't IGMP-join would go dark once any other node joins that universe.

1 — BigRack (192.168.0.254) — core, querier lives here

1a. Multicast core + querier

```
configure
ip igmp snooping
ip igmp snooping vlan 200
ip igmp snooping vlan 200 querier
ip igmp snooping vlan 200 querier version 3
ip igmp snooping vlan 200 immediate-leave
```

bridge multicast filtering is already in BigRack's config — verify, don't re-add. Then in the GUI set the querier's pace: **Multicast** → **IGMP Snooping** → **VLAN 200** → **Edit** → Query Interval **30**, MRouter Ports Auto Learn **enabled**. Verify

Multicast → **Unregistered Multicast** shows Forwarding on every port.

1b. QoS (4-queue box)

```
qos basic
qos trust dscp
qos map dscp-queue 56 to 4
qos map dscp-queue 46 to 3
qos map dscp-queue 8 to 2
```

Then GUI check **Quality of Service** → **General** → **Queue**: all queues Strict Priority (default, but confirm).

1c. Spanning tree — make BigRack the root

```
spanning-tree mode rstp
spanning-tree priority 4096
```

1d. Kill Auto Smartport, make the trunks static

The trunk allow-lists on gi45–47/49/50 exist only because dynamic Smartport macros applied them — dynamic macro config is REMOVED when a link drops. Make them static, then disable Auto Smartport:

```
interface gigabitethernet45
  switchport mode trunk
  switchport trunk allowed vlan add 100,200,300
exit
! repeat the same 3 lines for gi46, gi47, gi49, gi50
interface gigabitethernet48
  switchport access vlan 100
exit
no macro auto
```

If 'no macro auto' is rejected on this firmware, use the GUI: Smartport → Properties → Administrative Auto Smartport = Disable. gi48 ('Access') currently lands on VLAN 1 (management) — the line above parks it on Data instead. If it's truly unused, 'shutdown' is better.

2 — FOH (192.168.0.251)

2a. THE trunk fix — VERIFY FIRST

```
show interfaces switchport GigabitEthernet10
```

If the allowed list already shows 100,200,300 the backup was stale — skip. If it only shows VLAN 1, this uplink has been carrying management only:

```
configure
interface GigabitEthernet10
  switchport trunk allowed vlan add 100,200,300
exit
```

2b. Multicast

```
bridge multicast filtering
ip igmp snooping
ip igmp snooping vlan 200
no ip igmp snooping vlan 200 querier address
ip igmp snooping vlan 200 immediate-leave
```

The querier-address line in FOH's config was inert (querier never enabled) — clear it so nobody trips on it later. Verify Unregistered Multicast = Forwarding on all ports.

2c. QoS (8-queue box) + housekeeping

```
qos basic
qos trust dscp
qos map dscp-queue 56 to 8
```

```
qos map dscp-queue 46 to 7
qos map dscp-queue 8 to 2
spanning-tree mode rstp
no clock timezone
ip ssh server
```

FOH had 'no qos' — this turns QoS on properly. The 'clock timezone J 0' line was nonsense; clear it (set real time via GUI Administration → Time Settings while you're in there).

3 — Tech Table (192.168.0.253)

3a. Demote the querier (BigRack owns it now)

```
configure
ip igmp snooping
no ip igmp snooping vlan 200 querier
no ip igmp snooping vlan 200 querier address
ip igmp snooping vlan 200 querier election
ip igmp snooping vlan 200 immediate-leave
```

Tech Table was the only live querier — an edge switch, with election disabled and its management IP as querier address. The election line restores the default so it can never duel a future querier. Filtering + VLAN 200 snooping are already in its config — verify.

3b. QoS (8-queue box) + housekeeping

```
qos basic
qos trust dscp
qos map dscp-queue 56 to 8
qos map dscp-queue 46 to 7
qos map dscp-queue 8 to 2
spanning-tree mode rstp
ip ssh server
interface GigabitEthernet24
  switchport access vlan 100
exit
```

Tech Table had the contradictory 'no qos' + 'qos advanced-mode trust dscp' pair — 'qos basic' normalizes it. gi24 ('Emergency') currently sits on VLAN 1/management; the line above parks it on Data. gi19–23 are unconfigured — check the rack: anything unused gets 'shutdown'.

4 — Attic (192.168.0.252)

4a. Multicast (had none at all) + QoS (4-queue box)

```
configure
bridge multicast filtering
ip igmp snooping
ip igmp snooping vlan 200
ip igmp snooping vlan 200 immediate-leave
qos basic
qos trust dscp
qos map dscp-queue 56 to 4
qos map dscp-queue 46 to 3
qos map dscp-queue 8 to 2
spanning-tree mode rstp
```

4b. Credentials + banner — do this one carefully

Attic's config has NO local user account and its login banner literally says 'cisco' — it's advertising default credentials. Create the real account, log out, PROVE the new login works, then remove the default:

```
username brian password <strong-password> privilege 15
```

```
no banner login
no banner exec
! --- log out, log back in as brian, THEN: ---
no username cisco
```

Do not run 'no username cisco' until a login as brian has succeeded — locking yourself out of the Attic switch means a ladder and a paperclip.

4c. Naming + ports + trunk

```
interface vlan 100
  name Data
exit
interface gigabitethernet25
  switchport access vlan 100
exit
```

VLAN 100 was named 'Internet' here and 'Data' everywhere else — one name. gi25 (mode access, no VLAN) was parked on management; now Data (or 'shutdown' if unused). Then sort the uplink: run `show interfaces status` and find which port links toward BigRack gi49 ('Attic Trunk') — the backup suggests gi28 (Smartport-applied) while the static allow-list sits on gi27. Whichever port is the live uplink gets the static config:

```
interface gigabitethernet28  ! or gi27 — whichever is the live uplink
  switchport mode trunk
  switchport trunk allowed vlan add 100,200,300
exit
no macro auto
```

VLAN 101 (Paradigm) stays Attic-local on purpose — don't add it to the uplink unless Paradigm ever needs to leave the attic, in which case it also needs creating on BigRack and adding at both trunk ends. Attic firmware (1.3.5.58) is the oldest of the four — if any command is rejected, use the GUI equivalent.

5 — Verify (the part that makes it real)

1. On all four: `show ip igmp snooping vlan 200` — snooping operational everywhere; querier enabled on BigRack ONLY.
2. Wait ~2 minutes, then on BigRack: `show bridge multicast address-table vlan 200` — any multicast Dante flows appear as 239.255.x.x groups with member ports. A mostly-empty table on an all-unicast plant is normal.
3. Dante Controller on the Dante VLAN: every device visible, exactly ONE clock leader, no red clock events over a 10-minute watch. Pass audio on a test patch.
4. FOH: `show interfaces switchport GigabitEthernet10` now lists 1,100,200,300.
5. Optional but worth it on a dark day: reboot one edge switch and confirm its trunk comes back with all VLANs — that proves the Smartport removal took and the trunks are really static.
6. On all four: `copy running-config startup-config`, then download fresh backup files and replace the old ones.

Escape hatch: if one device loses clock or vanishes from Dante Controller after snooping goes live (a box that doesn't IGMP-join its groups), don't tear snooping out — set Multicast → Forward All → Static on just that device's port.

Parked for another day

Firmware: BigRack (1.4.8.6) and especially Attic (1.3.5.58) to final 1.4.x — do this on a separate dark day, one switch at a time, configs backed up first. BigRack's HTTPS certificate expired July 2018 — regenerate via Security → SSL Server. Standardize the 'brian' account on all four and retire 'cisco' everywhere. Edge-port/BPDU-guard on device-facing ports if STP ever gets touchy about hot-patching.